Orbit Determination: Practice Guide

AERSP 309, Kluever Ch. 3 · a guide to work through with the video

How to use it. The video pauses at each checkpoint. Work it here, on paper or a tablet, before you press play. Checkpoints 2.1, 3.1, 4.3(a) and 6.2(a) are answered by the video as soon as you resume. The others use a *practice orbit* that the video never solves, so you get real practice; check those against the full solutions on page 7. Stuck for five minutes? Read the hint on page 6.

The practice orbit. A tracking station measures, in the ECI frame,

$$\vec{r}_0 = (-5404.7,\ -9275.9,\ 2171.6)\ \text{km}, \qquad \vec{v}_0 = (3.4111,\ -2.2048,\ 4.8031)\ \text{km/s}.$$

Use $\mu = 3.986 \times 10^5$ km^3/s^2, $R = 6378.137$ km, $J_2 = 1.08263 \times 10^{-3}$. Carry four significant digits; your last digit may differ from the solutions by rounding.

Episode 2: Placing the Orbit in Space

Checkpoint 2.1 Inclination over ninety

video Ep 2, 2:51

An orbit has $i = 120°$. (a) At the ascending node, is the satellite crossing the equator heading north or heading south? (b) Is its motion over the Earth eastward or westward? (c) Does $\hat{h}$ point above or below the equatorial plane? Give a one-line reason for each.

Your answer

Episode 3: The Perifocal Frame

Checkpoint 3.1 **The third row of C_{PN}** video Ep 3, 1:56

With the course's rotations $C_1(x) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos x & \sin x \\ 0 & -\sin x & \cos x \end{bmatrix}$ and $C_3(x) = \begin{bmatrix} \cos x & \sin x & 0 \\ -\sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $C_{PN} = C_3(\omega)\, C_1(i)\, C_3(\Omega)$. Find the third row of C_{PN} in terms of i and Ω. Why does ω not appear in it?

Your answer

Episode 4: From $\vec{r}$ and $\vec{v}$ to the Elements

Checkpoint 4.1 **Practice orbit: a, $\vec{h}$, $\vec{e}$** video Ep 4, 2:07

For the practice orbit above, find (a) the energy $\mathcal{E} = v_0^2/2 - \mu/r_0$ and the semi-major axis a; (b) $\vec{h} = \vec{r}_0 \times \vec{v}_0$, one component at a time, and $h = |\vec{h}|$; (c) $\vec{e} = \dfrac{\vec{v}_0 \times \vec{h}}{\mu} - \dfrac{\vec{r}_0}{r_0}$ and $e = |\vec{e}|$; (d) the perigee and apogee radii. Check e a second way with $e = \sqrt{1 + 2\mathcal{E}h^2/\mu^2}$. Keep five digits in the two pieces of $\vec{e}$: its Y component is a small difference of nearly equal numbers.

Your answer

Checkpoint 4.2 **Practice orbit: C_{PN}, then i, Ω, ω** video Ep 4, 4:48

Using your $\vec{e}$ and $\vec{h}$ from 4.1, build $\hat{p} = \vec{e}/e$, $\hat{h} = \vec{h}/h$ and $\hat{q} = \hat{h} \times \hat{p}$, and stack them as the rows of C_{PN}. Then read off $i = \arccos C_{33}$, $\Omega = \operatorname{atan2}(C_{31}, -C_{32})$ and $\omega = \operatorname{atan2}(C_{13}, C_{23})$, each in $[0°, 360°)$. Confirm Ω and ω with the textbook route: $\vec{n} = \hat{K} \times \vec{h}$, $\cos\Omega = n_X/n$, $\cos\omega = \vec{n} \cdot \vec{e}/(n\,e)$. Arc cosine only returns 0° to 180°: if $n_Y < 0$, $\Omega = 360° - \arccos(n_X/n)$, and if $e_Z < 0$, $\omega = 360° - \arccos(\cdot)$. Which checks fire here?

Your answer

Checkpoint 4.3 **Which side of perigee?** video Ep 4, 5:11

(a) For the Molniya, $\cos\theta_0 = 0.1736$, so θ_0 is 80° or 280°. Predict: which quantity, computed straight from $\vec{r}_0$ and $\vec{v}_0$, decides between them, and how? (b) Find θ_0 for the practice orbit.

Your answer

Episode 5: Going Back

Checkpoint 5.1 **Practice orbit: back to $\vec{r}$ and $\vec{v}$** video Ep 5, 2:00

Take the practice orbit's elements $a = 12{,}000$ km, $e = 0.2$, $i = 50°$, $\Omega = 230°$, $\omega = 300°$, $\theta_0 = 75°$ (your answers to 4.1–4.3, so do those first). (a) Find $p = a(1 - e^2)$, $h = \sqrt{\mu p}$ and $r = p/(1 + e\cos\theta_0)$, then $\vec{r}_P$ and $\vec{v}_P$ in the perifocal frame. (b) Build $C_{PN} = C_3(\omega)C_1(i)C_3(\Omega)$ from the angles and compare it with your C_{PN} from 4.2. (c) Rotate back with $\vec{r}_N = C_{PN}^T\vec{r}_P$, $\vec{v}_N = C_{PN}^T\vec{v}_P$ and compare with the measured $\vec{r}_0$, $\vec{v}_0$.

Your answer

Episode 6: Earth Is Not a Sphere

Checkpoint 6.1 **Node drift** video Ep 6, 1:54

(a) For the practice orbit ($a = 12{,}000$ km, $e = 0.2$, $i = 50°$), find the mean motion $n = \sqrt{\mu/a^3}$ and the node drift $\dot{\Omega} = -\dfrac{3nJ_2R^2}{2a^2(1-e^2)^2}\cos i$ in degrees per day. Which way does the node move? (b) A circular orbit ($e = 0$, $a = R + 600$ km) must be sun-synchronous: $\dot{\Omega} = +360°$ per 365.24 days. What inclination does it need? (Put both rates in the same units first.)

Your answer

Checkpoint 6.2 **Where the perigee stops turning** video Ep 6, 2:32

$\dot{\omega} = -\dfrac{3nJ_2R^2}{2a^2(1-e^2)^2}\left(\frac{5}{2}\sin^2 i - 2\right)$. (a) Find every inclination in $[0°, 180°]$ where $\dot{\omega} = 0$. (b) Find $\dot{\omega}$ for the practice orbit in degrees per day. Does its perigee advance ($\dot{\omega} > 0$, turning with the motion) or regress ($\dot{\omega} < 0$)? Watch the two minus signs.

Your answer

Hints

Checkpoint 2.1 Which way the satellite crosses at the ascending node is part of the node's definition, not of i. For the motion, think about the extreme case $i = 180°$. For $\hat{h}$, look at its K component, $\cos i$.

Checkpoint 3.1 The third row of a product ABC is (third row of A) times BC. What is the third row of $C_3(\omega)$?

Checkpoint 4.1 Work $\vec{h}$ as $h_X = yv_z - zv_y$, $h_Y = zv_x - xv_z$, $h_Z = xv_y - yv_x$, and $\vec{v}_0 \times \vec{h}$ the same way. Your a should come out a round number.

Checkpoint 4.2 atan2 returns an angle in $(-180°, 180°]$; add $360°$ to a negative result. For the node route: arccos alone only reaches $0°$ to $180°$.

Checkpoint 4.3 Is the satellite moving away from Earth or toward it? Which half of the orbit does that put it in?

Checkpoint 5.1 Each column of C_{PN}^T is $\hat{p}$, $\hat{q}$, $\hat{h}$, so $\vec{r}_N = r_{P,x}\hat{p} + r_{P,y}\hat{q}$.

Checkpoint 6.1 Convert rad/s to degrees per day with $\times\, 180/\pi \times 86{,}400$. For (b), solve for $\cos i$.

Checkpoint 6.2 Set the bracket to zero and solve for $\sin i$; remember $\sin i = \sin(180° - i)$.

Solutions

Checkpoint 2.1 Inclination over ninety

(a) North. The ascending node is *defined* as the northbound equator crossing, whatever i is. (b) Westward. Below 90° the orbit runs with Earth's spin (prograde, eastward); past 90° it runs against it (retrograde); at $i = 180°$ it would run due west along the equator. (c) Below. $\hat{h} \cdot \hat{K} = \cos 120° = -0.5 < 0$, so $\hat{h}$ leans below the equatorial plane.

Checkpoint 3.1 The third row of C_{PN}

The third row of $C_3(\omega)$ is $(0, 0, 1)$, so the third row of the product is the third row of $C_1(i)C_3(\Omega)$, which is (third row of $C_1(i)$) $C_3(\Omega)$:

$$(0,\ -\sin i,\ \cos i)\begin{bmatrix} \cos\Omega & \sin\Omega & 0 \\ -\sin\Omega & \cos\Omega & 0 \\ 0 & 0 & 1 \end{bmatrix} = (\sin i \sin\Omega,\ -\sin i\cos\Omega,\ \cos i).$$

ω drops out because the last turn, $C_3(\omega)$, spins about $\hat{h}$ itself: it moves $\hat{p}$ and $\hat{q}$ but leaves the third row, $\hat{h}$, alone. Check: $C_{33} = \cos i = \hat{h} \cdot \hat{K}$, as in the picture.

Checkpoint 4.1 Practice orbit: a, $\vec{h}$, $\vec{e}$

(a) $r_0 = \sqrt{5404.7^2 + 9275.9^2 + 2171.6^2} = 10{,}953.0$ km and $v_0 = 6.2902$ km/s. $\mathcal{E} = 19.7833 - 36.3917 = -16.6085$ km^2/s^2, so $a = -\mu/(2\mathcal{E}) = 12{,}000$ km.

(b)

$$\begin{aligned} h_X &= (-9275.9)(4.8031) - (2171.6)(-2.2048) = -44{,}553.1 + 4{,}787.9 = -39{,}765 \\ h_Y &= (2171.6)(3.4111) - (-5404.7)(4.8031) = 7{,}407.5 + 25{,}959.3 = 33{,}367 \\ h_Z &= (-5404.7)(-2.2048) - (-9275.9)(3.4111) = 11{,}916.3 + 31{,}641.0 = 43{,}557 \end{aligned}$$

$\vec{h} = (-39{,}765,\ 33{,}367,\ 43{,}557)$ km^2/s, $h = 67{,}763$ km^2/s.

(c)

$$\begin{aligned} (\vec{v}_0 \times \vec{h})_X &= (-2.2048)(43{,}557) - (4.8031)(33{,}367) = -96{,}034.5 - 160{,}265.0 = -256{,}299.5 \\ (\vec{v}_0 \times \vec{h})_Y &= (4.8031)(-39{,}765) - (3.4111)(43{,}557) = -190{,}995.3 - 148{,}577.3 = -339{,}572.6 \\ (\vec{v}_0 \times \vec{h})_Z &= (3.4111)(33{,}367) - (-2.2048)(-39{,}765) = 113{,}818.2 - 87{,}673.9 = 26{,}144.3 \end{aligned}$$

Divided by μ: $(-0.64300,\ -0.85192,\ 0.06559)$. And $\vec{r}_0/r_0 = (-0.49344,\ -0.84688,\ 0.19826)$. So

$$\vec{e} = (-0.14956,\ -0.00504,\ -0.13268), \qquad e = \sqrt{0.14956^2 + 0.00504^2 + 0.13268^2} = 0.2000.$$

(d) $r_p = a(1-e) = 9{,}600$ km (altitude 3,222 km) and $r_a = a(1+e) = 14{,}400$ km. Check: $\sqrt{1 + 2(-16.6085)(67{,}763)^2/(398{,}600)^2} = 0.2000$. ✓

Checkpoint 4.2 Practice orbit: C_{PN}, then i, Ω, ω

$\hat{p} = \vec{e}/e = (-0.7478,\ -0.0252,\ -0.6634)$ and $\hat{h} = \vec{h}/h = (-0.5868,\ 0.4924,\ 0.6428)$.

$$q_X = (0.4924)(-0.6634) - (0.6428)(-0.0252) = -0.3267 + 0.0162 = -0.3105$$
$$q_Y = (0.6428)(-0.7478) - (-0.5868)(-0.6634) = -0.4807 - 0.3893 = -0.8700$$
$$q_Z = (-0.5868)(-0.0252) - (0.4924)(-0.7478) = 0.0148 + 0.3682 = 0.3830$$

$$C_{PN} = \begin{bmatrix} -0.7478 & -0.0252 & -0.6634 \\ -0.3105 & -0.8700 & 0.3830 \\ -0.5868 & 0.4924 & 0.6428 \end{bmatrix}$$

$i = \arccos(0.6428) = 50.0°$. $\Omega = \text{atan2}(-0.5868,\ -0.4924) = -130.0°$, so $\Omega = 230.0°$ (sine and cosine both negative: third quadrant). $\omega = \text{atan2}(-0.6634,\ 0.3830) = -60.0°$, so $\omega = 300.0°$ (fourth quadrant).

Textbook route: $\vec{n} = \hat{K} \times \vec{h} = (-h_Y,\ h_X,\ 0) = (-33{,}367,\ -39{,}765,\ 0)$, $n = 51{,}910$. $\cos\Omega = -33{,}367/51{,}910 = -0.6428$, and arccos gives 130°; but $n_Y < 0$, so $\Omega = 360° - 130° = 230°$. $\vec{n} \cdot \vec{e} = (-33{,}367)(-0.14956) + (-39{,}765)(-0.00504) + 0 = 4{,}990.4 + 200.4 = 5{,}190.8$, and $\cos\omega = 5{,}190.8/(51{,}910 \times 0.2000) = 0.5000$, so arccos gives 60°; but $e_Z < 0$ (perigee is below the equator), so $\omega = 300°$. Both routes agree. ✓

Checkpoint 4.3 Which side of perigee?

(a) The sign of $\vec{r}_0 \cdot \vec{v}_0 = r\dot{r}$. Negative means r is shrinking, so the satellite is falling toward perigee and θ_0 is in $(180°, 360°)$: 280° for the Molniya. Positive means it is climbing away from perigee and θ_0 is in $(0°, 180°)$.

(b)

$$\vec{e} \cdot \vec{r}_0 = (-0.14956)(-5404.7) + (-0.00504)(-9275.9) + (-0.13268)(2171.6)$$
$$= 808.33 + 46.75 - 288.13 = 566.95$$

and $e\,r_0 = 0.2000 \times 10{,}953.0 = 2{,}190.6$, so $\cos\theta_0 = 566.95/2{,}190.6 = 0.2588$ and arccos gives 75.0°.

$$\vec{r}_0 \cdot \vec{v}_0 = (-5404.7)(3.4111) + (-9275.9)(-2.2048) + (2171.6)(4.8031)$$
$$= -18{,}436.0 + 20{,}451.5 + 10{,}430.4 = 12{,}445.9 > 0,$$

so the satellite is climbing away from perigee and $\theta_0 = 75.0°$. (This is the opposite case to the Molniya.)

Checkpoint 5.1 Practice orbit: back to $\vec{r}$ and $\vec{v}$

(a) $p = a(1 - e^2) = 12{,}000(0.96) = 11{,}520$ km, $h = \sqrt{\mu p} = 67{,}763$ km^2/s, $r = p/(1 + e\cos\theta_0) = 11{,}520/1.05176 = 10{,}953.0$ km.

$$\vec{r}_P = (r\cos\theta_0,\ r\sin\theta_0,\ 0) = (2{,}834.9,\ 10{,}579.8,\ 0) \text{ km}$$

$$\vec{v}_P = \tfrac{\mu}{h}(-\sin\theta_0,\ e + \cos\theta_0,\ 0) = 5.8822\,(-0.9659,\ 0.4588,\ 0) = (-5.6818,\ 2.6989,\ 0) \text{ km/s}$$

(b) $C_1(50°)C_3(230°) = \begin{bmatrix} -0.6428 & -0.7660 & 0 \\ 0.4924 & -0.4132 & 0.7660 \\ -0.5868 & 0.4924 & 0.6428 \end{bmatrix}$; then $C_3(300°)$, with rows $(0.5,\ -0.8660,\ 0)$, $(0.8660,\ 0.5,\ 0)$, $(0, 0, 1)$, gives row 1 $= 0.5$(row 1) $- 0.8660$(row 2) $= (-0.7478,\ -0.0252,\ -0.6634)$ and row 2 $= 0.8660$(row 1) $+ 0.5$(row 2) $= (-0.3105,\ -0.8700,\ 0.3830)$; row 3 is unchanged. This is the C_{PN} of 4.2. ✓

(c) $\vec{r}_N = r_{P,x}\hat{p} + r_{P,y}\hat{q}$:

$$x : 2834.9(-0.7478) + 10579.8(-0.3105) = -2{,}119.9 - 3{,}285.0 \approx -5{,}404.9$$
$$y : 2834.9(-0.0252) + 10579.8(-0.8700) = -71.4 - 9{,}204.4 \approx -9{,}275.9$$
$$z : 2834.9(-0.6634) + 10579.8(0.3830) = -1{,}880.7 + 4{,}052.1 \approx 2{,}171.4$$

and the same with $\vec{v}_P$: $\vec{v}_N \approx (3.4108,\ -2.2049,\ 4.8030)$ km/s. Both match the measured $\vec{r}_0$, $\vec{v}_0$ to within the four-digit rounding of $\hat{p}$, $\hat{q}$. ✓

Checkpoint 6.1 Node drift

(a) $n = \sqrt{398{,}600/12{,}000^3} = 4.8028 \times 10^{-4}$ rad/s. The factor $\dfrac{3nJ_2R^2}{2a^2(1-e^2)^2} = 2.3908 \times 10^{-7}$ rad/s, so $\dot{\Omega} = -2.3908 \times 10^{-7} \cos 50^\circ = -1.5368 \times 10^{-7}$ rad/s $= -0.761^\circ$/day. Negative: prograde orbit, the node slides west.

(b) $a = 6378.137 + 600 = 6{,}978.1$ km, $n = 1.0831 \times 10^{-3}$ rad/s, and $\dfrac{3nJ_2R^2}{2a^2} = 1.4694 \times 10^{-6}$ rad/s $= 7.274^\circ$/day ($e = 0$). Setting $-7.274 \cos i = +0.9856$ gives $\cos i = -0.1355$, so $i = 97.79^\circ$. Check: between the video's 97.4° at 500 km and 98.2° at 700 km. ✓

Checkpoint 6.2 Where the perigee stops turning

(a) $\frac{5}{2}\sin^2 i = 2 \Rightarrow \sin^2 i = \frac{4}{5} \Rightarrow \sin i = \sqrt{4/5} = 0.8944 \Rightarrow i = 63.43^\circ$ or $180^\circ - 63.43^\circ = 116.57^\circ$.

(b) $\frac{5}{2}\sin^2 50^\circ - 2 = \frac{5}{2}(0.5868) - 2 = -0.5330$. With the factor from 6.1,

$$\dot{\omega} = -(2.3908 \times 10^{-7})(-0.5330) = +1.274 \times 10^{-7} \text{ rad/s} = +0.631^\circ\text{/day}.$$

Positive: $i = 50^\circ$ is below the critical inclination, so the perigee advances (turns in the direction of motion), just as the video's 50° example showed.